Raid AI at ArA-DF 2026: Data, Scale and Inference-Time Augmentation for Arabic Speech Deepfake Detection

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Abstract

We describe Raid AI's entry to Track 1 (Dialect Generalization) of the ArA-DF 2026 shared task on Arabic speech deepfake detection, which placed 1st of 15 teams at 0.8719% Equal Error Rate against a 14.54% official baseline. The configuration of the submitted system is proprietary and is not disclosed. We instead report the ablation behind it, re-scored on the official $144{,}210$-clip evaluation partition against the labels released after the challenge, with bootstrap confidence intervals and paired tests throughout. On development-test, corpus and augmentation alone move a fixed architecture from 13.13% to 1.37% EER. Encoder scale is not monotonic: a 1B encoder beats a 2B under a recipe matched on corpus, augmentation, learning rate, class weights and effective batch size, though the gap is inside the run-to-run spread we measure from three seeds. Three-crop test-time augmentation improves all ten models tested, by 17.4% relative on average, and is the only effect immune to seed variance because it is measured within a checkpoint; a per-crop decomposition shows the gain comes from averaging rather than from choosing a better window.

1 Introduction

Speech deepfake detection has advanced quickly for English, driven largely by the ASVspoof series (Yamagishi et al., 2021), but Arabic remains underrepresented despite a large speaker population and pronounced dialectal variation. ArA-DF 2026 addresses this gap. Track 1 asks whether a detector trained on one distribution of Arabic speech still works on dialects absent from training.

Our submission placed 1st of 15 teams, but the top four span 0.0546 percentage points and our margin over 2nd is 0.0088, so that ordering is not by itself informative. This paper reports the ablation behind the entry rather than the ranking: three encoder scales, five training corpora and two backends, including the runs that lost.

Two properties distinguish these measurements from the development-leaderboard feedback available during the challenge. The organisers released evaluation labels after it closed, so every model is re-scored on the official $144{,}210$-clip partition, and every comparison carries bootstrap intervals and paired tests, with run-to-run variance estimated from three seeds. Several differences that look meaningful as point estimates do not survive that treatment, and we mark them.

Related work. Self-supervised encoders are the standard anti-spoofing frontend: wav2vec 2.0 (Baevski et al., 2020) and its cross-lingual extension XLS-R (Babu et al., 2022). AASIST (Jung et al., 2022) and attentive statistics pooling (Okabe et al., 2018) are common backends, and the ArA-DF baseline pairs XLS-R 300M with AASIST. Codec and transmission augmentation was central to ASVspoof 2021 (Yamagishi et al., 2021), whose condition sets we reuse, and MLAAD (Müller et al., 2024) supplies multilingual spoofed speech. Shared tooling has emerged to make such results comparable: AUDDT (Zhu et al., 2025) puts 31 audio deepfake datasets behind one benchmark interface, and DeepFense (El Kheir et al., 2026), in which our models are implemented, standardises architectures, losses and augmentation over 100 recipes and reports from 400+ models that curated training data drives cross-domain generalization. Lin et al. (2026) lead the 14-set Speech DF Arena benchmark (Dowerah et al., 2025) with a Conformer encoder, crediting multi-source data, balanced sampling and augmentation rather than architecture; our N-ALL result qualifies this, since adding out-of-domain data *without* rebalancing hurt. Multi-crop evaluation is established in vision (Krizhevsky et al., 2012) but is rarely ablated in anti-spoofing; Section 5.3 measures it here.

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2 Task Setup

Track 1. Systems perform binary bonafide-versus-spoof classification on Arabic speech, ranked by Equal Error Rate (EER), the task's only metric; the evaluation emphasises dialects absent from training.

Partitions and baseline. One hidden test set is split into two *disjoint* partitions, with no clip in both: the development phase scored 16,023 clips (25.57% bonafide), 10% of the test set, and the evaluation phase the remaining 144,210 (25.59% bonafide). The evaluation phase permitted three submissions per team, best taken as the official score. The organisers' baseline, XLS-R 300M with an AASIST backend, scores 14.65% on development-test and 14.54% on the evaluation partition. We train the same architecture in Table 3, which separates the contribution of our data pipeline from that of the architecture.

Scores. Submissions are `audio_id,logit` pairs with higher values indicating bonafide. We emit the log-likelihood ratio $\mathrm{LLR} = z_{\text{bona}} - z_{\text{spoof}}$ from the classifier logits. EER is invariant under monotonic transforms, so it needs no calibration.

3 Models and Training Corpora

Architecture. Every model shares a three-stage design: a self-supervised encoder, a backend, and a two-class head. We use XLS-R (Babu et al., 2022) at 300M (315,438,720 parameters), 1B (962,497,408) and 2B (2,159,259,648), each fully fine-tuned. Inputs are 64,600 samples at 16 kHz (4.04 s); shorter clips are repeat-tiled rather than zero-padded, because leading-silence duration is a documented shortcut in anti-spoofing corpora (Müller et al., 2021) and zero-padding manufactures that cue. The default backend is temporal-channel modelling (Truong et al., 2024) with attentive statistics pooling (Okabe et al., 2018) under inverse-frequency weighted cross-entropy; one model uses an AASIST backend (Jung et al., 2022) on the same frontend.

Codec-simulation augmentation. Each training clip is re-rendered offline under a set of transmission conditions following the ASVspoof 2021 definitions (Yamagishi et al., 2021). The **narrow** set has the seven LA telephony conditions: pass-through, A-law, μ-law, G.722, GSM full-rate, Opus, and a PSTN approximation (300–3400 Hz band-pass, then 8 kHz A-law, then GSM). The **broad** set adds the nine DF compression conditions: its own pass-through, MP3 and AAC at 32 and 128 kbps, Ogg Vorbis at $q0$ and $q6$, and two double round-trips (MP3 to AAC, Ogg to AAC). Sixteen in total, two of them uncompressed. Training also applies room impulse response convolution and MUSAN (Snyder et al., 2015) additive noise (SNR 0–20 dB) on the fly, at p=0.25 each for narrow-set runs and p=0.5 each for broad-set runs. Table 1 lists the corpora. Narrow-to-broad is therefore a regime change, not a controlled variable.

Corpus	Sources	Cond.	Rows	Bona.
N-AR	AR	7	157,500	33.3%
N-AR+EN	AR, EN	7	335,160	21.1%
N-ALL	AR, EN, ML	7	961,275	7.3%
B-AR	AR	16	360,000	33.3%
B-AR+EN	AR, EN	16	766,080	21.1%

Table 1: Training corpora. AR is the ArA-DF training split (22,500 clips), EN is ASVspoof2019 (Todisco et al., 2019) train, ML is MLAAD (Müller et al., 2024). N and B denote the narrow (7) and broad (16) condition sets. Rows is sources × conditions.

Submitted system. Our official Track 1 submission, the Raid AI model, is built from the components and training data above. Its configuration is proprietary and is not disclosed here; it is not one of the models in Table 3 and is not reproducible from this paper. Of the axes that table reports we can place it on one: the three-crop test-time augmentation of Section 5.3 was applied to it. Limitations names the four we do not answer.

4 Experimental Setup

Training. All models are fully fine-tuned with Adam at 1×10^{-6} and exponential decay (γ=0.9) in bfloat16 autocast, with inverse-frequency class weights per corpus. Effective batch size is 64 for every run, over 8×A100-80GB for the 2B models (per-GPU batch 2, accumulation 4), 8×A100-40GB for the 1B (4, 2) and 4×L40S for the 300M (16, 1); the AASIST model uses one L40S (16, 4). Software is PyTorch 2.6.0. Checkpoints were selected per run on the development leaderboard, so the selected epoch differs across rows and is reported for each. Epoch budgets were also not fixed: under a bounded compute allowance we extended only the configurations that looked promising early, so a row's epoch count partly reflects its own early results (see Limitations).

Rank	Team	EER (%)
1	**Raid AI (ours)**	**0.8719**
2	haitao_xu	0.8807
3	hassan111	0.8917
4	pi-labs.ai	0.9265
	official baseline	14.5400

Table 2: Official ArA-DF 2026 Track 1 evaluation-phase leaderboard, top four of fifteen teams, with the organisers' baseline. Evaluation partition, 144,210 clips. EER is the task's only ranking metric; against the released labels our submission also reaches 99.13% accuracy and 98.86 macro-F_1 at its own EER threshold, and 79.85%/78.05 at τ=0.

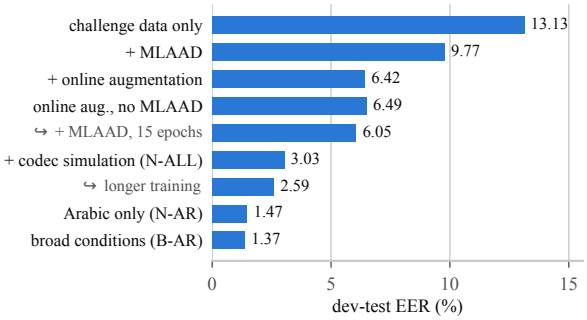


Figure 1: One 300M encoder across successive training regimes, development-test EER, all post-hoc. Grey subrows are variants of the row above; corpora are defined in Table 1.

Scoring protocol. The leaderboard result in Table 2 is the only figure scored during the challenge. The rest is computed afterwards by re-scoring stored predictions against the released labels; re-scoring our winning submission with the same code recovers the official 0.8719% to four decimals, which validates the procedure. Every evaluation-partition figure for a model, at one crop and at three, comes from one inference pass, so the two are paired by construction.

Uncertainty. We report two variance sources separately. *Evaluation variance* is a percentile bootstrap over clips: 2,000 resamples of the evaluation partition with replacement, reporting the 2.5/97.5 percentiles. Comparisons use a paired bootstrap, scoring both systems on the same resample so that between-clip variance cancels. *Training variance* is estimated from three runs of one configuration (300M on N-AR) differing only in random seed, which give 1.47, 1.85 and 1.93% on development-test: an unrounded standard deviation of 0.25 and range of 0.47 points. Training variance is therefore the larger of the two.

5 Results

5.1 Training Regime

Figure 1 tracks a fixed 300M architecture across training regimes. Most of these models predate the evaluation phase and were never scored on it, so this comparison is development-test throughout.

The span is an order of magnitude, 13.13% to 1.37%, with the same encoder, backend and optimiser. Adding MLAAD to the challenge data removes a quarter of the error and online room-impulse and noise augmentation a further third. Codec simulation then halves what remains (6.05 to 3.03%), the largest single effect we measured. All four steps exceed the 0.47-point seed range of Section 4.

The lower bars qualify the value of data volume. Dropping MLAAD and ASVspoof2019 from the codec-simulated corpus, a reduction from 961,275 rows to 157,500, improves EER from 2.59% to 1.47%. Out-of-domain data was harmful at this stage. Class balance is a partial confound, since the larger corpus is 7.3% bonafide against 33.3%, and we did not separate the two.

5.2 Encoder Scale

Table 3 gives the model comparison on the evaluation partition. Every model improves on the 14.54% baseline by at least five times, the seven strongest by roughly an order of magnitude. The AASIST row shares the baseline's architecture exactly and scores 2.05% on the same partition, so the data pipeline rather than the backend accounts for most of the gain.

A 1B beat a 2B. The 1B and 2B rows on N-AR share corpus, augmentation, learning rate, class weights and effective batch size. The 1B wins by 0.18 points, and the paired bootstrap interval $[-0.22, -0.11]$ excludes zero. The effect is nonetheless smaller than the 0.47-point seed range, and the two selected checkpoints sit at different epochs (10 against 11), so we report it as a counterexample to defaulting to the largest encoder rather than as a property of the architectures.

Scale and augmentation are of similar size. Holding the corpus at N-AR and growing the encoder from 300M to 2B is worth 0.10 points $[+0.04, +0.18]$; broadening the augmentation regime at 300M is worth 0.10 $[+0.05, +0.17]$.

			EER (%)			
Encoder	Corpus	Ep.	Dev	Eval	3-crop	Δ
2B	B-AR	9	1.07	**1.19**	1.00	−0.19
2B	B-AR+EN	5	1.22	1.21	**0.96**	−0.26
1B	N-AR	10	1.20	1.25	1.12	−0.14
300M	N-AR+EN	10	1.27	1.37	1.14	−0.22
300M	B-AR	8	1.37	1.42	1.14	−0.28
2B	N-AR	11	1.39	1.43	1.12	−0.31
300M	N-AR	17	1.47	1.53	1.18	−0.34
300M	N-AR†	18	1.93	1.95	1.56	−0.39
300M	N-AR‡	17	2.12	2.05	1.61	−0.44
300M	N-ALL	7	2.59	2.78	2.66	−0.12

Table 3: Single-model EER, all rows post-hoc, sorted by Eval. Dev is development-test (16,023 clips); Eval and 3-crop are the evaluation partition (144,210 clips) scored at one and at three crops in the same inference pass, and Δ is their difference computed before rounding. Ep. is the selected epoch. †: a second random seed (240) of the row above. ‡: AASIST backend in place of TCM. Bootstrap 95% intervals on the Eval column are ±0.06 to ±0.09 points, and ±0.14 on the N-ALL row.

Both exclude zero and are the same size, so the data does not rank the two interventions. It does support treating a 300M encoder under the broad regime and a 2B under the narrow one as indistinguishable (-0.01, $[-0.07, +0.06]$) at $7\times$ fewer parameters.

Adding ASVspoof2019 to B-AR likewise more than doubles the corpus and moves the 2B by $+0.02$ points $[-0.03, +0.08]$, the wrong way and within noise: no measurable benefit from out-of-domain English on a dialect-generalization track.

5.3 Test-Time Augmentation

Each clip is scored under three deterministic windows instead of one, at offsets 0, 0.5 and 1 of the slack between the clip length and the 4.04 s window: head-aligned, centred and tail-aligned. Clips shorter than the window are repeat-tiled as at training time, so their crops coincide. The three log-likelihood ratios are averaged; nothing is retrained.

The gain is significant for all ten models, ranging from 0.12 to 0.44 points (17.4% relative on average, mean 0.27 points). Every paired interval excludes zero. Because both scores come from the same checkpoint, training variance cancels, which makes this the only comparison in the paper not confounded by the seed spread of Section 4.

The per-crop scores decompose the effect. The head crop is the best single crop for all ten models, at 1.62% mean EER, while the centre and tail crops are worse at 1.83% each. The mean of the three nevertheless beats every individual crop for every model, reaching 1.35%: two individually weaker views still help. Section 5.4 confirms the mechanism.

Duration	Clips	Bona.	1 crop	3 crops	Δ
< 4.04 s	11,488	18.1%	5.67	5.67	+0.00
4.04–8 s	45,614	24.2%	1.56	1.43	−0.13
8–12 s	55,940	25.0%	1.23	0.86	−0.37
≥ 12 s	31,168	31.4%	1.08	0.64	−0.44

Table 4: EER (%) by clip duration on the evaluation partition, averaged over the ten models of Table 3. The analysis window is 4.04 s.

Averaging two of the three crops places a point between. Averaged over the two pairs that contain it, the head crop plus one other gives 1.40%, so the second crop buys 82% of the total gain and the third the remaining 18%, and three beat that two-crop average for all ten models. Returns diminish but had not stopped by three; we did not run more.

5.4 Error Analysis

The released metadata carries per-clip duration, which partitions the evaluation set along the axis our window length acts on (Table 4).

Errors concentrate on short utterances. Clips below the window are 8.0% of the set but carry an EER of 5.67%, three to five times the rate elsewhere, and are the most spoof-heavy bucket. This is the dominant residual error mode for every model we trained, and a property of the input rather than of any architecture.

The table also isolates the test-time augmentation mechanism. Below the window the crops coincide and the method cannot act: the gain is exactly zero for all ten models. Above it the gain grows monotonically with slack, 0.13 to 0.44 points. Averaging predicts this; a better-window account does not, since a longer clip gives no reason for one window to be better.

6 Conclusion

Our submission placed first on ArA-DF 2026 Track 1, 0.8719% EER against a 14.54% baseline. The system itself is not disclosed; the ablation behind it is. Data and augmentation moved a fixed architecture by an order of magnitude, encoder scale was not monotonic, and three-crop test-time augmentation improved all ten models at no cost.

Limitations

The submitted system is not described. The configuration of the Raid AI model is proprietary. Components, corpora and augmentation are reported in full, but the submitted system cannot be reproduced from this paper and is not among the models tabulated. Of the five axes of Table 3 we report one, that three-crop test-time augmentation was applied to it. We do not report its encoder scale, its backend, its training corpus or its augmentation regime, because answering any of them would disclose how the system is built. A reader can therefore locate the submitted system on one axis of the ablation and on none of the other four, and the leaderboard result and the ablation study should be read as separate contributions.

Training variance is estimated from one configuration. Our seed spread comes from three runs of the 300M model on N-AR, measured on development-test. We assume it transfers to the 1B and 2B rows and to the evaluation partition, which we did not verify; retraining the larger encoders across seeds was beyond our compute budget. Differences smaller than 0.47 points should be read with that in mind.

Epoch budgets were allocated adaptively, not fixed. Training the 2B models is expensive, so we could not give every configuration the same number of epochs. We ran each configuration for a few epochs, inspected the development leaderboard, and spent the remaining budget on the ones that looked promising, stopping the others early to free capacity for further experiments. The epoch counts in Table 3 are therefore an outcome of that triage rather than a controlled setting, and they correlate with how well a configuration was already doing. A row that trained longer had earned the right to; a row that stopped early may not have converged. Differences between rows consequently confound encoder, corpus and optimisation budget, and none of our comparisons should be read as holding training length constant.

Uncontrolled confounds. Checkpoints were selected per run on the development leaderboard, so the epoch differs across rows, and several runs were stopped before the epoch budget completed. Expanding the codec condition set also expands the corpus and coincided with a higher on-the-fly augmentation rate. The corpora differ in class balance, which we did not separate from domain in the N-ALL result.

Development-leaderboard selection. Every challenge decision came from the public development leaderboard, a disjoint partition 11% the size of the evaluation one, so model selection could have fitted it. Across the ten models the two partitions correlate at Spearman $\rho{=}0.99$, with development EER optimistic by 0.06 points, so the effect appears small here; we did not reserve a clean partition to bound it.

Effect sizes sit close to the noise floor. Evaluation intervals are ±0.06 to ±0.09 points wide and the seed standard deviation is 0.25. The training-regime steps of Figure 1 and the test-time augmentation gains clear both; the scale and augmentation comparisons of Section 5 clear only the first. Our leaderboard margin of 0.0088 clears neither, which is why we regard our result as competitive with rather than superior to the systems behind it.

The analysis window was never varied. Short clips are the dominant error mode, and a shorter or duration-adaptive window is the obvious response. We did not test one. The $4.04\,$s window is inherited from the pretrained encoder's own training recipe, and changing it means retraining every model rather than re-scoring stored predictions. The crop count stops at three for a different reason: a fourth or fifth crop needs a fresh inference pass over the evaluation partition for every model, whereas the two-crop point of Section 5.3 is a subset of the three we already had.

Re-scoring is not bit-reproducible. Under bfloat16 autocast a repeated inference pass does not reproduce its own scores: for three of the seven models we scored twice, 55–62% of clips moved, by a mean 0.03 log-likelihood-ratio points, changing EER by at most 0.01. Every evaluation-partition figure for a model therefore comes from one pass, so that its one-crop and three-crop columns are paired.

Post-hoc scoring. All figures except the leaderboard result were computed after the challenge closed against released labels. They had no bearing on the standing, and the model selection that produced the submission was made without them.

Scope, and why calibration is absent. We report Track 1 only and rank by EER, the task's

metric. EER is threshold-free and says nothing about whether a deployed operating point would be well calibrated. We did not investigate calibration: no calibration set was released, the challenge score did not depend on it, and our models emit an uncalibrated LLR. The cost is visible in the caption of Table 2, where the submitted system reaches 99.13% accuracy at its own EER threshold and 79.85% at $\tau{=}0$, a 19-point gap that is entirely threshold placement. Error at a fixed threshold is also asymmetric: at $\tau{=}0$ the three strongest models of Table 3 reject 0.5–0.6% of bonafide clips while accepting 1.9–2.6% of spoofs, so residual error is dominated by spoofs passing as genuine. Every such rate describes an uncalibrated operating point.

Ethics Statement

Detecting synthetic speech is a defensive application, but the ablation that shows what makes a detector strong also shows where detectors are weak, and we report those weaknesses rather than only the win. Error on utterances shorter than the 4.04 s analysis window is three to five times the rate elsewhere, and a system deployed at a fixed threshold instead of one fitted on labelled data gives up nineteen points of accuracy. An attacker can measure both properties from any deployed system, so stating them is more useful to a defender than withholding them is protective.

All data is the organisers' ArA-DF 2026 release together with ASVspoof2019 (Todisco et al., 2019), MLAAD (Müller et al., 2024) and MUSAN (Snyder et al., 2015), used under their published terms. We collected no speech, and none of our analysis identifies a speaker. Track 1 is a dialect-generalization task, so error rates can differ by dialect; the released labels do not carry dialect, we did not measure per-dialect error, and that gap should be closed before a system of this kind informs a decision about a speaker. A detector of this accuracy is a filter, not evidence, and the accuracy is conditional on a fitted threshold: at $\tau{=}0$ the strongest models of Table 3 still accept about one spoofed clip in fifty, and the submitted system, whose scores are uncalibrated, accepts more than one in four.

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